

AGCS
Ground Control System.

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asseco

Asseco Ground Control Software.

Asseco Ground Control Software is the main component of Asseco Ground Control Systems dedicated for integration with any unmanned platform, used for complete management of tasks before, during and after the mission.

The modular structure of the system allows for the creation of various software configurations. AGCS is offered in both a standardized and a “tailor-made” version, which is prepared for specific customer requirements.

AGCS Functionalities

Checking the elevation of the area
Checking the height of flight route points in relation to the topography of the area. Advance notifications in the case where the planned flight route is in collision with a terrain obstacle.

Map sheets
Switch between layers with map sheets. Multiple layers can be used at the same time and their transparency can be adjusted.

2D/3D view
Free switching from diagonal to flat and vice versa.

UAV tracking
Enable/disable map tracking for the UAV object.

UAV track
Delete a track mark behind a UAV object.

Basic parameter tag
Enable/disable the information marker of the UAV object.

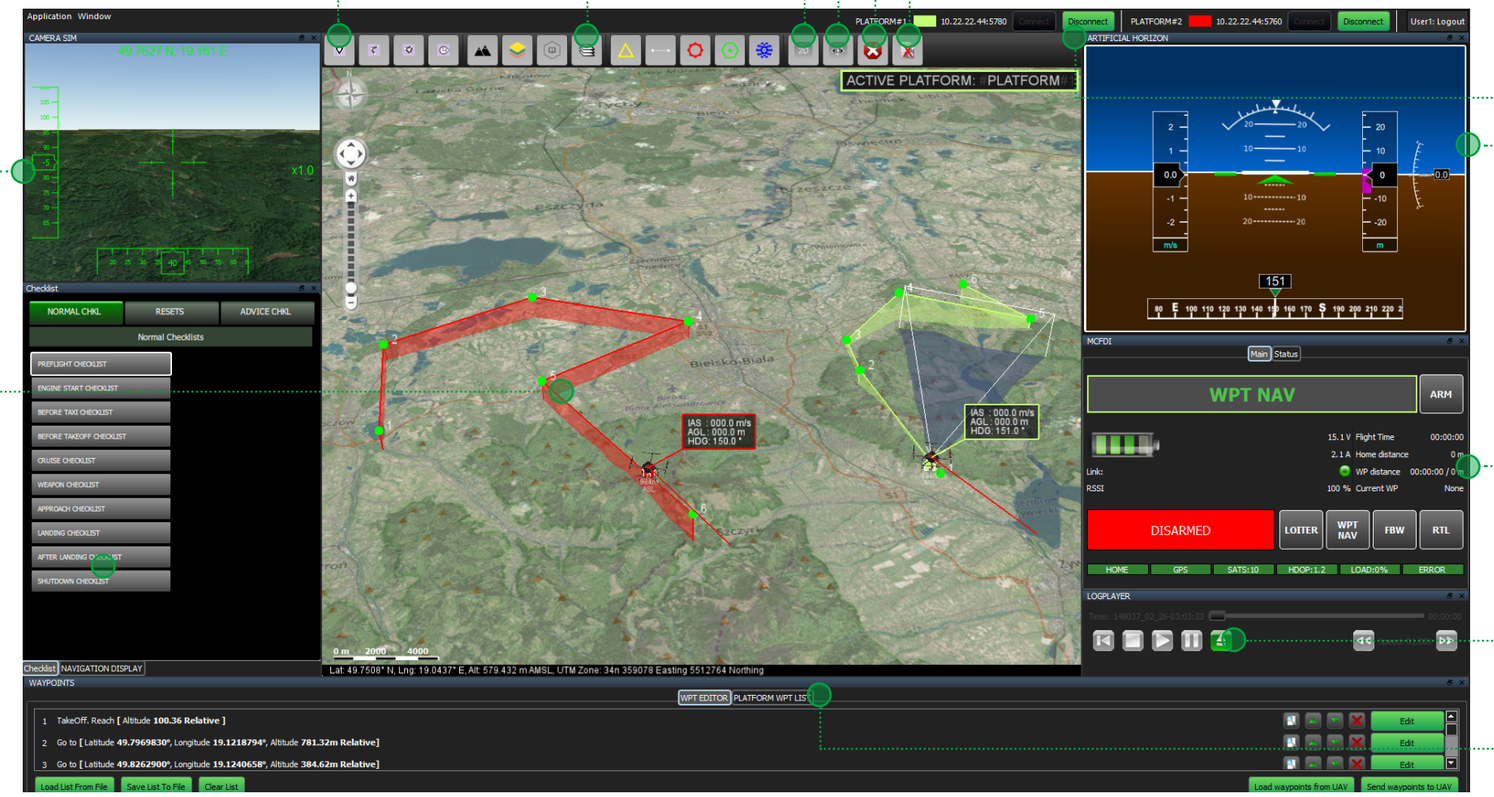
Map tools
Adding markers, tracks, areas or ellipses.

Sensor view
Adjustment of the size of the window with the displayed image as needed. Switching between views from individual sensors or a stream from both at the same time.

Map tools
Visualization of the sensor’s direction and field of view directly on the map sheet.

Checklist
A systemic way to check the UAV system before performing a mission.

Supported Map Formats
WGS84, MGRS, DTED, GeoTif,



Switching between platforms
Control up to three platforms during one mission.

Artificial horizon module
Free switching between sensor view, 3D terrain model, or MALE-type platform-tailored view.

UCFDI (UAV Computer Flight Data Indicator)
Visualization of information about currently enabled flight modes and data concerning system operation.

Logplayer
Reconstructing the course of completed missions.

Modifiable user interface layout
Drag&drop placement of individual module windows in the interface according to user preferences.

View of the range of the communication system
Information for the operator regarding the UAV staying within the communication range

Photogrammetric flight path generator
Automatic generation of the flight path of a photogrammetric mission on the basis of preset parameters.

Mission planning module
A planning tool for autonomously executed missions. Possibility of changing navigation points during the mission.

Why AGCS?

Security
Original security architecture. Full ownership and control of source code. The software development process is carried out in accordance with CMMI-DEV Level 3.
The use of AGCS guarantees the users full control and security of the processed data.

Simulation environment
An integrated simulation environment allows training for UAV operators using real software and the hardware part of the ground control station. The air platform model is implement-ed on the basis of the actual characteristics of the simulated platform.

Intuitiveness
User interface designed in cooperation with UAV operators.

Easy Integration
STANAG compliance and full control over a dedicated platform management module (VSM).

Personalization
Easily customize and save the layout and settings of the graphical interface according to your preferences.

Versatility
It can be installed on any device (smartphone, tablet, laptop, PC).
Supported operating systems:
Windows, Linux, Android, iOS.

Interoperability
Cooperation with external IT systems i.a. for command, mission management, reporting and fleet maintenance.

Key benefits.

Platform manufacturers:
Using AGCS, the manufacturer gains high quality, proven software that is the central element of the unmanned aerial system. The software can be adapted to the specifics of the integrated set and to the air traffic rules and regulations currently in force.
Thanks to the implementation of AGCS, the manufacturer reduces expenditures on configuration and integration of sensors or open-source software. In this way, they gain a better cost-effect ratio and the manufacturer's participation in the integration is limited to the necessary minimum.

End users:
AGCS is also a great offer for individual users. The consumer version is fully compatible with the Mavlink communication protocol, so it can be used in systems equipped with autopilots from the PX4 family.
The implementation of AGCS allows for the handling of different BSPs by means of a single ground-based control station software. The customer gains the possibility to unify the process of operator training.

AGCS – Extensions.

Airspace Awareness Module
A module ensuring reception of signals from transponders mounted on manned aircraft. It allows visualization of civil air traffic directly in AGCS. It provides visualization of airspace zones. It enables real-time monitoring of traffic within 300 km of the ground control station, without the need for an Internet connection.

RT module
A hardware real-time module that enables reliable operation of a ground control station in the event of critical operating system or computer failure.



Features and benefits of the system.

Wide range of applications
AGCS universal software is designed to be used in systems of various designs (fixed-wing, VTOL) and classes (micro, mini, tactical and MALE).
With the support of civil aviation standards and compliance with STANAG standarization documents, it enables the implementa-tion of missions in both civil and military applications.

Compliance with standards
The AGCS version for uniformed services complies with NATO standards for data exchange and processing. It meets the following standards:
STANAG 4586 – Architecture and interoperability, STANAG 4607 – KLV meta-data processing and STANAG 4671 – Airworthiness.

What do we offer?

AGCS + automatic control system

A complete „Plug and fly" solution for your platform. A dedicated, platform manufacturer-specific solution consisting of AGCS and an automatic control system installed on an air platform. This service integrates on-board equipment and provides support for recognition sensors (e.g. intelligent target tracking functions). Software-adapted set for the specific requirements of the integrated system. Possibility of integration of any sensors and actuators, modification of control rights. Comprehensive integration service.

Integration of unmanned systems (design and implementation):

- navigation and communication equipment, intelligence systems and automatic control systems,
- training systems and simulators,
- hardware control stations.

Trainings:

- dedicated training in the use and maintenance of unmanned systems in the performance of tasks as part of the business operations of a given organization with the use of Asseco Ground Control Software,
- operation and execution of unmanned aerial vehicle missions with the use of Asseco Ground Control Software.



Bet on proven software.

2014
Start of work

2016
first shipment for the Polish Armed Forces

2016
AGCS in the project EDA - ERA (Enhanced RPAS Automation)

2018
First shipment to Africa (Nigeria)

2019
AGCS + UAV MayFly for the Polish Armed Forces

Asseco
Ground Control Station.

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